Non-Invasive Methods for PWV Measurement in Blood Vessel Stiffness Assessment

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(Methodological Review)

***Abstract*—In recent years, statistical studies highlighted an increasing incidence of cardiovascular diseases (CVD) which reflected on additional costs on the healthcare systems worldwide. Pulse wave velocity (PWV) measurement is commonly considered a CVD predictor factor as well as a marker of Arterial Stiffness (AS) since it is closely related to the mechanical characteristics of the arterial wall. An increase in PWV is due to a more rigid arterial system. Because of the prevalence of the elastic component, in young people the PWV is lower than in the elderly. Nowadays, invasive and non-invasive methods for PWV assessment are employed: there is an increasing attention in the development of non-invasive devices which mostly perform a regional PWV measurement (over a long arterial portion) rather than local (over a short arterial portion). The accepted gold-standard for non-invasive AS measurement is the carotid-femoral PWV used to evaluate the arterial damage, the corresponding cardiovascular risk and to adapt the proper therapy. This review article considers the main commercially available devices underlining their operating principles in terms of sensors, execution mode, pulse waveforms acquired, site of measurement, distance and time estimation methods, as well as their main limitations in clinical practice.**

***Index Terms*—Arterial stiffness, CVD, non-invasive commercial devices, PWV measurement.**

I. Introduction

PULSE Wave Velocity (PWV) measurement is nowadays used as a marker of Arterial Stiffness (AS) and therefore it can be related to different cardiovascular diseases (CVD). In particular, the AS is usually used to predict the cardiovascular and renal failures, assess hypertension (i.e., increase in arterial pressure), coronary artery disease and diabetes [1], [2], identifying patients at risk at an early stage [3]. By the term arterial stiffness, it is typically meant a generic parameter which is indirectly estimated from PWV measurements. As a consequence, it depends on the site in which such measurements are performed. It coincides with aortic stiffness only in case of invasive PWV assessment on the aorta. Since in the period 1990-2015 the number of CVD cases has dramatically increased in Europe, prediction becomes an element of primary importance. According to the data reported from Global Burden of Disease project, 10.6 million (5.1 million men and 5.5 million women) and 11.3 million (about 5.4 million men and 5.8 million women) new cases of CVD have been reported in Europe in 2010 and 2015, respectively [4]. According to some studies by the Department of Public Health, University of Oxford, the costs of hypertension treatments and, in general, of CVD are very high in many European countries and the overall CVD is estimated to cost € 210 billion a year in EU [4]. In Table I, the costs of CVD in Italy, Germany, France, U.K. and Spain in 2015 are compared (€ thousands). Among them, Italy and Germany have the highest total health care costs for CVD. Another study reports an estimation of healthcare costs attributable to hypertension on a large North American population (about 2 millions of adults) [5]: it is highlighted that 10.2% of direct healthcare spending could be attributed to hypertension, with similar estimates in USA (12%). Furthermore, the study suggests that hypertension healthcare costs could increase by about 95% from 2010 to 2020 in Alberta. Therefore, prevention of hypertension and CVD, could limit this cost growth.

A further aspect to be considered is the reduced efficacy of anti-hypertensive therapy because of several factors (e.g., patient beliefs and lifestyle, health-literacy, socio-economic status) and their relationship with the healthcare system. Such aspect may affect the effectiveness of anti-hypertensive treatments [6], [7]. This could represent a further source of expense for different countries. Mennini *et al*. [8] have reported these costs and also how in five European countries the anti-hypertensive therapy impacted economically speaking.

In sight of these data, the increasing interest in the Pulse Wave Velocity (PWV) assessment for CVD prediction is fully justified. Both invasive (single-catheter or double-catheter procedures) and non- invasive methods can be employed for PWV assessment, even though nowadays, the development of non-invasive devices in the clinical practice is strongly recommended to overcome the intrinsic complications related to the invasive approaches. In particular, these devices are currently able to perform a regional (over a long arterial portion) or local (over a short

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TABLE I
COST OF CARDIOVASCULAR DISEASES IN SOME EUROPEAN COUNTRIES (2015)

Country	Primary care [G€]	Medication [G€]	Total health care costs [G€]	Cost per capita [€]
Italy	0.8	4.4	15.7	258
Germany	3.2	5.3	28.3	349
France	0.6	3.8	15.2	229
UK	1.6	2.7	12.3	190
Spain	1.3	3.5	9.2	199

arterial portion) PWV measurement. Such measurements can be carried out with different approaches that have been analysed throughout this work. Among them, the accepted gold-standard for non-invasive AS measurement is the carotid-femoral PWV [1]. The purpose of this review article in such a context is to analyse the different techniques for PWV assessment with particular reference to non-invasive ones. Therefore, the main commercially available devices (Arteriograph, Complior, Mobil-O-Graph, MRI, pOpmètre, PulsePen, SphygmoCor, Ultrasound systems, VP1000) have been investigated from different points of view: employed sensors, execution mode, pulse waveforms acquired, measurement site, distance and time estimation methods and main limitations in clinical practice.

II. ARTERIAL STIFFNESS AND PULSE WAVE VELOCITY

The mechanical properties of the wall of the large arteries depend on their viscoelastic characteristics and influence both the diastolic and systolic blood pressure fluctuations around the mean pressure values, as well as the type of flow in the large caliber arteries [9], [10]. These properties are usually described in terms of compliance, distensibility and stiffness.

Volumetric compliance C is defined as a change in volume ΔV, related to the change in inner pressure ΔP [10], [11]:

$$C = \frac{\Delta V}{\Delta P} \tag{1}$$

The primary effect of compliance is to damp pressure and volume fluctuations due to the discontinuous action of the left ventricle. On the other hand, the compliance is related to vessel capability of storing the pulsatile energy in the systole and returning it to during the diastole.

Distensibility χ, is a measure of the arterial capability to expand and contract with cardiac pulsation and relaxation [12]. There is a large agreement on the mathematical definition of the volumetric distensibility, that can be defined as the variation in volume ΔV, related to the pressure variation ΔP as a function of the initial volume V of the vessel [11], [13]:

$$\chi = \frac{\Delta V}{\Delta P \cdot V} \tag{2}$$

Alternatively, the volume variation ΔV can be derived from the lumen area stroke change ΔA or from the artery diameter change between systole and diastole [1]. Therefore, (2) can be written as:

$$\chi = \frac{\Delta A}{\Delta P \cdot A} = \frac{d_S^2 - d_D^2}{\Delta P \cdot d_D^2} \tag{3}$$

where A is the initial vessel cross-section, while d_S and d_D are the systolic and diastolic arterial diameters, respectively.

Arterial stiffness is inversely related to distensibility χ [11], [14] and indicates a lowering of arterial compliance, or a change in vessel wall properties, or both. To this regard, a general agreement on its mathematical definition seems not to be reached yet, while scientific literature usually relies on a different parameter [15] known as beta stiffness index (β) and defined as:

$$\beta = \frac{\ln\left(\frac{P_S}{P_D}\right)}{\varepsilon} = \frac{\ln\left(\frac{P_S}{P_D}\right)}{\frac{d_S - d_D}{d_D}} \tag{4}$$

where P_S and P_D are the systolic and diastolic pressures respectively and ε is the arterial strain. Therefore, all the above three parameters C, χ and β are pressure-dependent and are all related to each other.

In the study of the wall mechanical characteristics, the incremental elastic modulus E_{inc} is another quantity usually taken into account: it is a measurement of an elastic material stiffness [11], [14] and characterizes the elastic properties of the structural components independently from the geometry of the vessel [16], [17]. Anyway, since the above quantities are difficult to evaluate in vivo and usually need invasive methods, PWV is usually measured and considered between the most common and reliable indicators of arterial stiffness. In particular, PWV is defined as the velocity of propagation of the pressure waves, generated by the systolic contraction of the heart, along the arterial tree [18]: it is an index of the AS and viscoelastic properties of the arterial wall [16] as the velocity of propagation of the arterial wave is closely linked to the properties of the tunica media and to the elastic modulus of the wall [17], [19]; therefore, in young subjects who have a prevalent elastic component, the wave propagation speed is slower than in elderly patients [9], [20]. In healthy conditions, the pulse wave propagates into the aorta at a speed of about 5 m/s and increases progressively in small arteries up to 10-15 m/s [10], [21], as shown in Fig. 1. When the normal muscle-elastic wall structure is altered (e.g., due to pathologies), the viscoelastic properties are partially lost, leading to an increased AS [22]. Therefore, such change leads the system of large arteries to become less efficient in damping the

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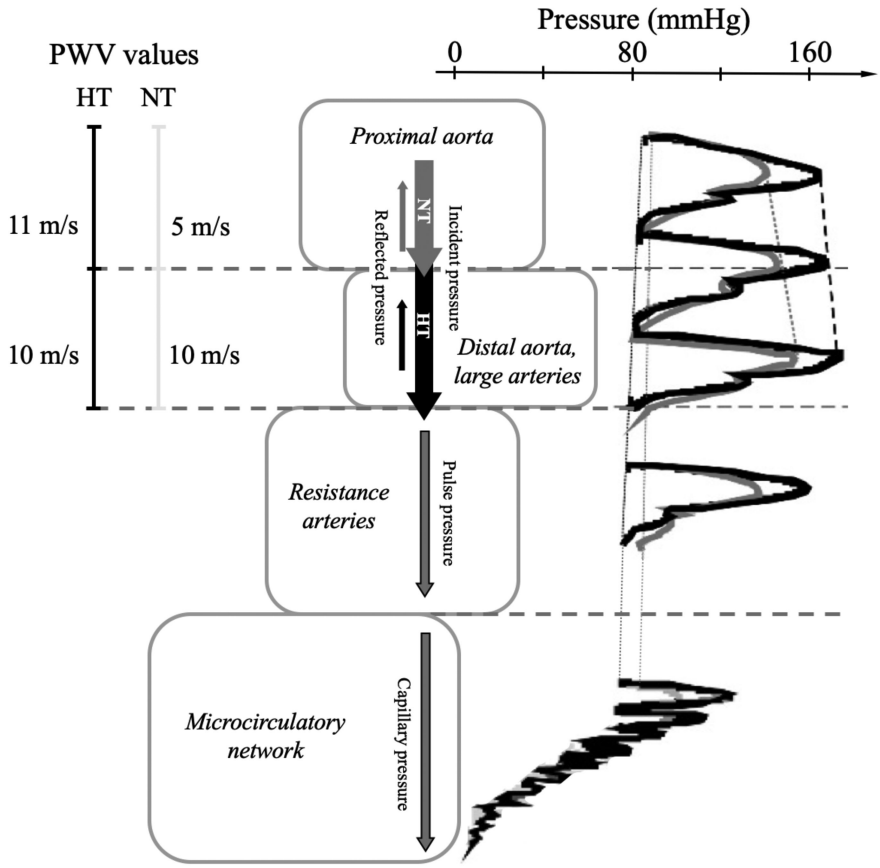


Fig. 1. Pressure wave reflection and impedance mismatch; adapted from [37]. In normotension (NT), the proximal aorta is more distensible than in the distal aorta, therefore an arterial stiffness gradient occurs, with the aortic PWV lower than the distal aorta PWV. The gradient can be considered an impedance mismatch that generates wave reflections returning at low PWV to the aorta during the diastole phase. In this case, since the reflected wave arrives quite late in systole, there is little superposition of reflected waves to incident waves, therefore systolic blood pressure (SBP) is not increased. In hypertension (HT), the stiffness gradient tends to disappear or is inverted (aortic PWV is more than peripheral PWV, reducing the impedance mismatch), the pulsatile pressure is insufficiently dampened and damages the microcirculation. Moreover, reflected pressure waves travel backward at higher velocity than in the NT case, therefore they arrive at the central aorta in early systole and are superimposed on the incident pressure wave: an increase of the SBP occurs.

systolic-diastolic pressures and in converting the flow from pulsatile to continuous. In these conditions, the emptying of the left ventricle is hindered by the increase of arterial impedance and post-load. As a consequence, an increase of cardiac work can be observed with remodeling and hypertrophy of the left ventricle, as well as the alteration of coronary perfusion [23]–[25]. From a pathophysiologic point of view, to allow the ejection of the systolic volume in a more rigid arterial system, it is necessary to generate a higher systolic arterial pressure. Moreover, the alterations of viscoelastic wall properties modify the conduction capacity of the pulse wave generated by the ventricular systole. In a more rigid arterial system, the pulse wave propagates faster in a similar way to the reflected wave at the bifurcations level [17], [26]: the shape of the pulse wave and the increase of peak pressure into the arterial vessels are determined by the contribution of both the incident and reflected pressure waves. For some pathologies (i.e., arteriosclerosis, diabetes) the PWV can reach up to 15 m/s or more [27], [28].

In clinical research, the *propagative model* [27], [29], is used to represent the arterial tree. The model consists of a viscoelastic tube terminating with the peripheral resistance, therefore offering distributed elastic properties able to generate a pressure wave traveling along the tube. In particular, the Moens-Korteweg equation is usually used to calculate the PWV in an infinitely long, straight, isolated and cylindrical thin-walled tube with elastic, isotropic and homogenous walls and filled with a homogenous incompressible and non-viscous liquid [30]. The velocity of the arterial wall wave is given by the following relationship [31], [32]:

$$PWV = \sqrt{\frac{E_{inc} \cdot h}{\rho \cdot d}} \tag{5}$$

where E_{inc} is Young's elastic modulus of the wall, h is the wall thickness, ρ is the density of the liquid and d the diameter of the tube. The Bramwell-Hill equation was derived from the Moens-Korteweg equation to describe the relationship between the vascular wall distensibility χ and PWV [33]. In this equation, the PWV intrinsically varies with blood pressure, and it is directly proportional to the arterial stiffness:

$$PWV = \sqrt{\frac{\Delta P \cdot V}{\Delta V \cdot \rho}} = \sqrt{\frac{1}{\chi \cdot \rho}} \tag{6}$$

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TABLE II
PWV ACCORDING TO THE MEASUREMENT SITE

PWV	Measurement site
aoPWV	Aortic PWV
cfPWV	Carotid-femoral PWV
baPWV	Brachial-ankle PWV
ftPWV	Finger-toe PWV

Therefore, considering (6) and the inverse relation between AS and χ, a direct relationship between PWV and AS can be assessed: an increase of PWV is strongly related to an increased AS.

The *local PWV* measurement is usually performed on a short portion (few centimeters) of artery, and it is considered a useful tool for the early diagnosis of the local stiffness increment in the arterial wall [34], [35]. It provides diagnostic information on biomechanics of local arterial wall [36].

On the other hand, the *regional PWV* is usually done in two different arteries and only provides the average PWV over a long segment (about 50-60 cm) along which mechanical characteristics vary [18], [27]. Therefore, biomechanical properties of a small portion cannot be achieved by regional PWV measurements so preventing the localization of arterial abnormalities [36].

AS has been widely employed in research protocols, and its use as a prognostic marker in clinical practice is still uncommon [3]. It's debated whether AS increase is a cause or an effect of hypertension, and whether large or small arteries are damaged first [37]. A wide range of pulse signals can be used to assess PWV, such as pressure waves [38], [39], distension waves [40] or flow waves. Even if these waves have not a common nature, they could be interchangeable because, theoretically, at the beginning of the cardiac cycle they are in phase. From the comparison between pressure and distension waveforms, it can be highlighted that pressure waves are slightly less smoothed than the distension ones [41]. Once applied on the artery of interest, the pressure transducer could possibly provide distorted waveforms due to an improper application [41]. Nevertheless, the pressure wave is the most used signal.

The PWV can be derived from measurements of the pressure wave on two sites of the arterial tree [42]: a proximal point (e.g., carotid, brachial arteries) and a distal or peripheral point (e.g., femoral, ankle arteries). A list of the measurement sites that are generally used in clinical practice is reported in Table II. The time delay with which the pressure wave is acquired in the proximal portion with respect to the distal may be measured directly on the patient or calculated from other quantities. Therefore, PWV is evaluated by the following relationship:

$$PWV = \frac{L}{\Delta t}\,[m/s] \tag{7}$$

where L is the *travel distance*, usually measured on the skin between two recording sites and Δt is the *transit time* of the arterial pulse along the analyzed arterial portion, i.e., the time delay of the pressure wave from the first to the second recording site.

III. INVASIVE PWV ASSESSMENT METHODS

Invasive *aortic PWV* (aoPWV) measurements are performed on patients who undergo routine coronary angiography using standard, fluid-filled, pigtail catheters [43]. The protocols may imply both single-catheter or double-catheter procedures.

The catheter position is checked through a fluoroscopy procedure [44], pulse pressure wave signals are recorded and afterwards, a tape is used to mark the positions at the catheter. After removing the catheter, the invasive travel distance (i.e., the catheter length) is measured between the two marked points and finally, transit time is estimated [44]. Since the invasive method shows several drawbacks, its use can be limited only to cardiac and/or vascular diagnostic and procedures [45].

The development of non-invasive devices to assess PWV and AS in the clinical practice is strongly recommended. This in order to overcome all the problems associated with invasive procedures, such as risks associated with comorbidity (e.g., coexistence of several diseases in the same patient), necessary presence of a surgery physician, high-dependency on the operator-skill and lack of comfortability for the patient.

IV. NON-INVASIVE PWV ASSESSMENT METHODS

The simplest, non-invasive, robust and reproducible method to determine AS is the measurement of PWV [1], [42]. Several methods to estimate local PWV are proposed in literature [46], including the wave intensity analysis [47], the flow/area (Q/A) method [48], the diameter/flow velocity (ln(D) U-loop) method [49], and the gradient method [50]. The gold-standard for non-invasive AS measurement is the *carotid-femoral PWV* (cfPWV), that is related to the extensively accepted propagative model of the vascular system [1] and has been successfully validated as an independent predictor for all-cause mortality and cardiovascular morbidity [51], [52]. Therefore, the determination of cfPWV has been implemented in the European Society of Cardiology (ESC)/European Society of Hypertension (ESH) Guidelines for the Management of Arterial Hypertension [2] to evaluate arterial damage, the corresponding cardiovascular risk and to adapt the proper therapy.

In 2000, an alternative technique for the assessment of aortic stiffness has been developed in Japan [53] through the brachial-ankle (baPWV). Compared to cfPWV, the baPWV seems to require a simpler and faster recording procedure, which is less traumatic for the patients, since its estimation can be carried out using pressure cuffs wrapped around upper arms and ankles [54]. Nevertheless, there are several criticisms on baPWV measurement [55]. Scientific literature pointed out two main issues [56]:

- in the same arterial tree, the pulse wave does not move directly from the brachial to the post-tibial arteries;
- since brachial and post-tibial arteries can be considered as peripheral arteries, then baPWV may be an index of peripheral rather than central AS.

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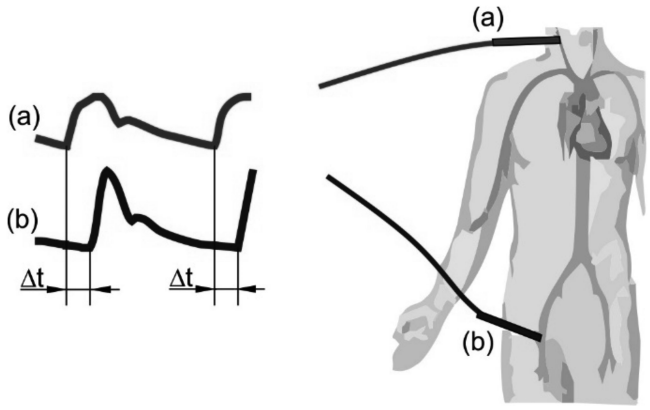


Fig. 2. One step measurement of carotid-femoral PWV; adapted from [42].

In clinical practice and in scientific literature, the cfPWV is frequently indicated as aoPWV [44], because the aging vascular deteriorations mostly arise along the aorta, while such phenomena are less significant in muscular arteries [57] and therefore they don't often give a contribution in the prognosis [58]. However, on closer examination cfPWV reveals itself as a mere surrogate of aoPWV, since some differences between the two measurements can be detected [59], [60]. Furthermore, carotid and femoral pressure waveforms could be difficult to acquire in patients with metabolic syndrome, obesity, diabetes or peripheral artery disease [47], [61]. Nowadays, it is impossible to directly measure aoPWV with non-invasive methods, excepted for phase-contrast magnetic resonance imaging (MRI) [59], [62].

The most common regional PWV measurement is carried out where the pulsation can be easily felt: on the right common carotid artery (proximal point) and on the right femoral artery (distal or peripheral point). While the pulse wave transit time can be usually measured in an accurate way, the distance assessment is much less accurate [63]. Some of the main recommendations regarding the acquisition procedures of cfPWV are summarized as follows [1], [64]:

- perform measurements in supine position after at least 10 min of rest;
- repeat measurements at the same time of the day to avoid diurnal variations;
- obtain data during at least one respiratory cycle (about 5-6 s);
- evaluate the mean of at least two measurements;
- do not perform the measurement of cfPWV in some pathological cases (e.g., arrhythmia, unstable clinical situation).

The exam can be carried out in two different ways [42], [65]:

1) *in one step* (Fig. 2), using two probes simultaneously (tonometers, ultrasound probes, oscillometers or mechanoreceptors). The time delay Δt is usually measured between the feet of the two waveforms (*foot-to-foot velocity* method).
2) *in two steps* (Fig. 3), using a single probe combined with an electrocardiographic trace. In this way, the PWV can be estimated in two steps in rapid succession, having as reference point the QRS wave of the ECG, that allows the synchronization of the two pulse wave times [66]. Transit time Δt can be retrieved from the difference between the delays T_2 and T_1, of the distal pulse wave and the proximal one, respectively, with respect to R-wave.

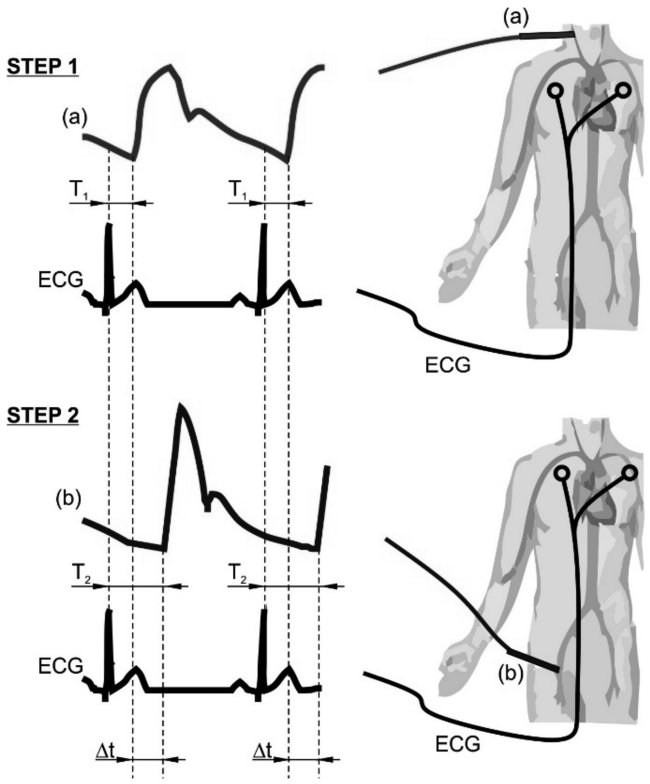


Fig. 3. Two steps measurement of carotid-femoral PWV; adapted from [42].

Results are comparable to the previous method, nevertheless a constant heart rate must be assumed between the acquisition of the proximal pulse wave and the acquisition of the peripheral one. Owing to this requirement, it is mandatory to measure arterial pressure and heart rate at the same time as arterial pulse transit time. PWV measurements for which the difference between the average values of the cardiac period (i.e., RR interval) recorded in the carotid artery and in the peripheral artery are greater than 10% are unreliable [42].

V. Identification of the Wave Foot

Since the pressure waveform does not repeat identically through time because of inter- and intra-individual variability, a key point is the calculation of the transit time from a reference part of the waveform. In particular, because of reflections, the pressure wave shape is given by the sum of the incident wave, due to the heart contraction, and the reflected wave, from the peripheral part of the arterial system, therefore, the wave shape changes according to the amplitude and timing variations of the reflected wave, which in turn vary with the measurement site. This results in a systematic difference between the minimum and maximum values of the pressure waves in one site and the ones found on other sites [67]. Consequently, reference points must be

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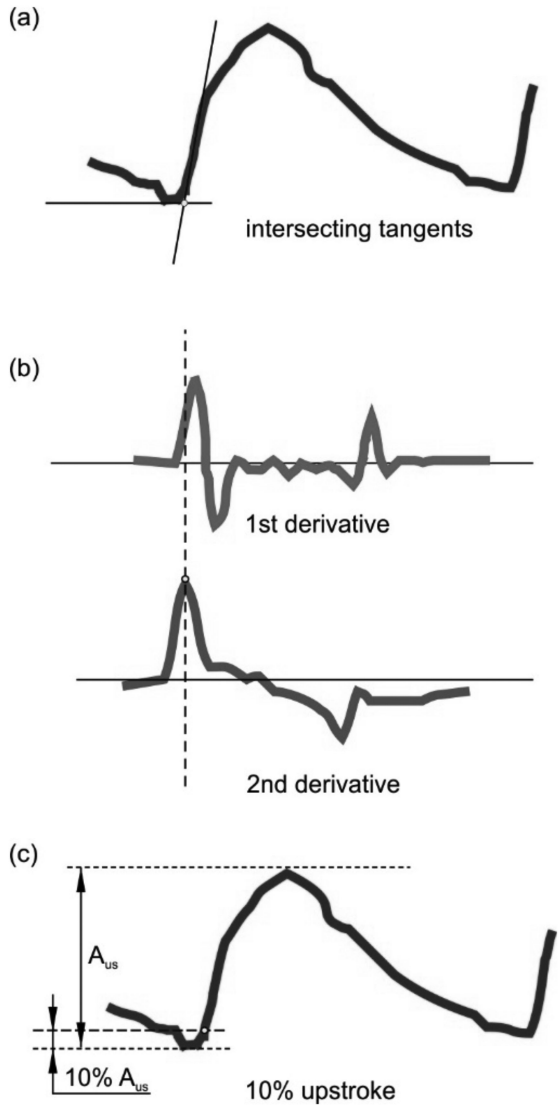


Fig. 4. Identification of the foot of the pulse wave (here the distension waveform of the carotid artery is represented); adapted from [41].

selected on the waveform independently from the measurement site. Since systolic upstroke seems well-defined for all sites, it is usually used as the reference point, even if it should be taken into account that the wave foot enlargements show a mild transition from diastole to systole and, especially in correspondence of the carotid site, the early wave reflections amounts provide a non-negligible contribution to the wave frequency content [41], [68]. The identification of the wave foot is usually achieved by one of the following algorithms [41], [69], [70] (Fig. 4):

A) intersecting tangents, defined by the intersection between the tangent through the steepest part of the pressure-wave upstroke and the point of lowest pressure;
B) maximum upstroke of the second derivative;
C) 10% of the distance between maximum and minimum points of the pulse pressure waveform.

Intersecting tangent method was proven to be the most reliable [67], even if the other methods provide compatible results and they are not affected by the heart rate. By the way, according to

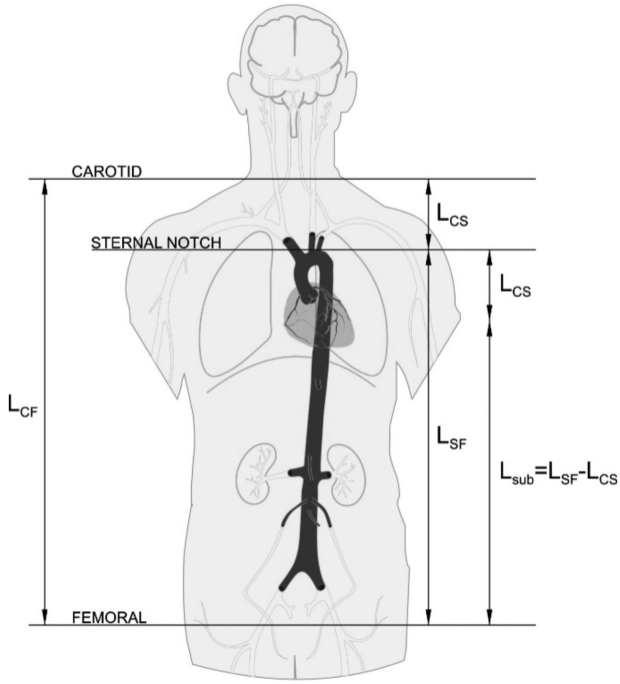


Fig. 5. Direct and subtractive method to measure the carotid-femoral path length; adapted from [41].

[46], it should be pointed out that tracking the exactly matched characteristic features (homologous points) based on fiduciary time-points is impractical. Thus, the true pulse transit time determination over a regional segment of the artery remains ambiguous if the same fiducial time-point identification algorithm is applied on proximal and distal pulses [46].

VI. Sites and Distance Measurement

In literature, various methods have been proposed for the estimation of the travel distance length [64], [66], [71], as reported in Table III. The simplest way to assess the pathway is the *direct method*, where a transcutaneous measurement is performed by means of a tape meter or a caliper (L_{CS} in Fig. 5) on the body surface; in that case, the coarse approximation on the travel distance produces a systematic overestimation that influences the prognostic significance of cfPWV [72], [73], e.g., the above estimation may even exceed 25% of the true value [41], [64].

An alternative way to determine the travelled distance is the so-called *subtractive method* [74], [75]. An example is shown in Fig. 5, where the pathway is assessed by subtracting L_{CS} (i.e., the distance between the carotid site and the sternal notch) from L_{SF} (i.e., the distance between the sternal notch and the femoral site). The study in [64] shows that this method underestimates the real travel distance by 29.2% (when the sternal notch is used as reference point) and 10.3% (when the suprasternal notch is used as landmark). Currently, the best approximation of the true travelled aortic path length [71] is the 80% of the direct carotid-femoral distance, i.e., 0.8 times the distance from the femoral artery, measured on the patient's skin by tape or caliper, to the carotid artery. This distance seems the most accurate

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TABLE III
MAIN METHODS PROPOSED FOR THE PATHWAY MEASUREMENT IN PWV ASSESSMENT

Method	Pathway	Path length [cm]	Estimation error [%]
Direct	L_{CF}	63.6 ± 4.4	25.4
Subtractive	$L_{SF} - L_{CS}$	35.9 ± 4.3	29.2
	$L_{SSF} - L_{CSS}$	45.5 ± 4.5	10.3
Direct 80%	$0.8 \times L_{CF}$	50.9 ± 3.5	0.4
Weber	BH/4 + K	50.2 ± 2.3	1.0

CF carotid-femoral; SF sternal notch-femoral; CS carotid-sternal notch; SSF suprasternal notch-femoral; CS carotid-suprasternal notch; BH body height. Measurements are reported as mean ± SD. The path length and the estimation error values are taken from [63].

among all other estimations, only slightly overestimating the actual travelled distance by 0.4% [43], [64], [76]. Furthermore, it is of primary importance that the distance tape measurement is carried out in a straight line and when such issue cannot be respected (e.g., severe obesity), an infantometer may be used upside-down to make a distance measurement with higher reliability [64].

Another method to determine the travel distance, introduced by Weber *et al*. [75], is based on the body height, as expressed in the following relationship:

$$\text{Travel Distance} = \frac{\text{Body height}}{4} + K \tag{8}$$

where K is an empirical constant connected to calibration (i.e., 7.28) [74]. The equation (8) has been validated and confirmed as the second-best method [64], [71] in travel distance evaluation, despite the underestimation of about 1%.

In Table III a comparison is proposed among the most common methods to evaluate the travel distance [64]: the direct 80% and Weber method are the most accurate, with an estimation error up to 1%. Relative uncertainties increase dramatically to about 30% for the other methods proposed in the scientific literature.

VII. AVAILABLE SYSTEMS FOR NON-INVASIVE PWV ASSESSMENT

The non-invasive PWV measurement can be accomplished by a set of dedicated customized systems. Among them, some are routinely used in diagnostic clinic environment while other ones are not so widespread, as image-based screening tools (e.g., MRI [27] and ultrasound systems [77]). These tools are often seen as expensive and difficult techniques. On the other side, there are some commercially available systems that are quite widespread and able to detect several hemodynamic parameters. Among them, the most common in the market are listed here in alphabetical order [1], [27], [65], [66], [78], [79], while their characteristics are summarized in Table IV:

- *Arteriograph*® (TensioMed, Hungary). The PWV is estimated from a single-site by an oscillometric brachial-cuff, recording the supra-systolic waveform at the brachial artery site (Fig. 6). In particular, it is based on the measurement of the time distance between the first wave peak (left ventricular ejection) and the second one (assumed to be due only to the incident pressure wave reflection from the aorto-iliac bifurcation), with a further subtraction of the brachial artery transit time. The cuff pressure is set at least 35 mmHg over the real systolic one [80], [81]. The distance from the upper edge of the pubic bone to the sternal notch is used [43], [82] (jugulum-symphysis, it is usually about 50 cm). When the artery is occluded and the flow is stopped, the acquired waveforms are pure pressure waves because in such situation the brachial

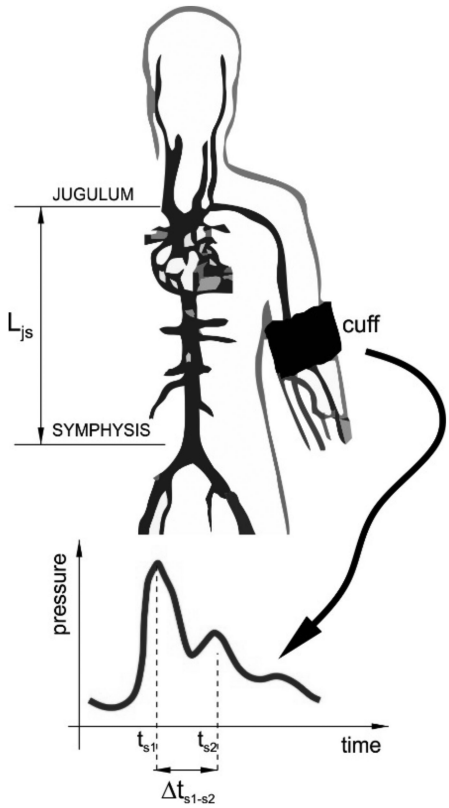


Fig. 6. Operating principle of Arteriograph. The PWV is calculated from PWV = $2 \cdot L_{js} / \Delta t_{s1\text{-}s2}$; adapted from [80].

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TABLE IV
AVAILABLE SYSTEMS FOR NON-INVASIVE PWV ASSESSMENT

System		PWV	Execution mode	Pulse waveform	Site	Distance estimation method	Time estimation method	Sensor/ Method
Arteriograph®		Regional	One step	Pressure	Jugulum-symphysis	Direct	Peak to peak	Arm pressure cuff
CAVI-VaSera®		Regional	One step	Pressure	Brachial-ankle	Direct	Foot-to-foot	Pressure cuffs (4), microphone
Complior®		Regional	One step	Pressure	Carotid-femoral, Carotid-brachial	Direct	2nd derivative	Piezoelectric mechano-transducers
Mobil-O-Graph®		Regional	One step	Pressure	Jugulum-symphysis	Direct	4th derivative	Arm pressure cuff
MRI		Local	One step	Flow	Aortic arch	–	Intersecting tangent	Phase-contrast
NiHem®		Regional	Two steps	Pressure	Carotid-femoral, Carotid-radial	Direct	–	Applanation tonometry
pOpmètre®		Regional	One step	Pressure	Finger-toe	Based on body height	Foot-to-foot	Pletismography
PulsePen®		Regional	One or two steps	Pressure	Carotid-femoral	Subtractive (suprasternal notch)	Intersecting tangent	Applanation tonometry
Sphygmo Cor®	CvMS	Regional	Two steps	Pressure	Carotid-femoral	Subtractive (suprasternal notch)	Intersecting tangent	Applanation tonometry
	XCEL		One step	Pressure and Volume				Tonometer and thigh cuff
US systems	Echo-tracking: HDI-lab, Philips; E-Traking, Aloka; RF-QAS, Esaote	Regional	One or two steps	Flow, Volume or Distension	Carotid-femoral, Regional carotid PWV, Regional aortic PWV, Aorta-femoral	–	–	Echo tracking; Ultrasound medical imaging; Doppler probes and ECG
	A- B- M-Mode	Local	–	Volume or Distension	Ascending aorta	–	–	US probe
	Doppler	Regional/ Local	One or two steps	Flow	Carotid-femoral	Subtractive (suprasternal notch)	Foot-to-foot	Doppler probe
Vicorder®		Regional	One step	Pressure	Carotid-femoral, Carotid-brachial	Direct	2nd derivative	Neck pad (partial cuff) bladder-type thigh/arm cuff
VP1000®		Regional	One step	Pressure	Brachial-ankle arteries	Weber	Sharp systolic upstroke	Pressure cuffs

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artery wall is almost still under the cuff [43]. In [83], the fundamental operating principle of the device seems to be validated. Nevertheless, the measurement accuracy of the PWV is considered challenging, due to the difficulty in assessing the exact location of the reflected wave. In [43], [84], a good agreement with invasive methods (considered as reference) has been found with a relative difference between about 4% up to 20% of the reference value.

- *Complior*® (ALAM Medical, France). The system simultaneously records the arterial wave pulses by means of up to 4 piezoelectric transducers applied directly to the skin (carotid, femoral, radial distal arteries) [89]. The transit time is estimated between each simultaneous recorded wave through the application of the second derivative algorithm in correspondence of the wave foot. On the other hand, the distance travelled by the pulse wave is measured between the two recording sites (carotid-femoral, carotid-brachial or femoral-dorsalis pedis) by the direct method [90]–[92]. This device is considered one of the most reliable and accurate systems in the market for non-invasive PWV assessment [77]. Its accuracy depends mainly on the travel distance measurement, since the pulse transition time evaluated by the method has an accuracy equivalent to the one provided by catheter methods [93]. In [88] the mean difference ± SD in PWV estimation between Complior-Analyse and the reference invasive method was -0.73 ± 2.83 m/s ($r = 0.64$).
- *Mobil-O-Graph*® (IEM GmbH, Germany). The device is similar to Arteriograph, as it is based on an oscillometric brachial-cuff, embedded with a high-fidelity pressure sensor, that allows the recording of the brachial pulse waveform (BPW). From the acquired wave, a 3 steps algorithm is applied to reconstruct the aortic pulse waveform (AOW) by means of a generalized transfer function [85], [86]. In other terms, the system assumes that AOW can be obtained through a frequency content modification of the BPW; to this aim the generalized transfer function is applied to BPW. After that, AOW is split into incident and reflected pulse waves; the time difference between the forward and reflected waves is used to estimate the PWV value [87]. In a recent study [88], the mean difference ± SD in PWV estimation between Mobil-O-Graph and the reference invasive method (i.e., FS-Stiffcath catheter) was -1.01 ± 2.54 m/s ($r = 0.71$), showing a weaker agreement with invasive PWV respect to cfPWV estimations from other devices.
- *MRI* has the advantage of measuring with high accuracy travel distance length of pulse waves in the proximal and distal aorta, despite the tortuous nature of the vessel [78], [94] without geometrical assumptions. Phase-contrast MRI (PC-MRI) allows to assess blood flow velocity for each anatomic plane [78]. The pulse wave arrival time is determined from the intersection between the waveform baseline and the upstroke tangent [95]. Nevertheless, MRI is an expensive and non-portable system, affected by poorer time resolution [62], and may be uncomfortable for patients who suffer from claustrophobia.

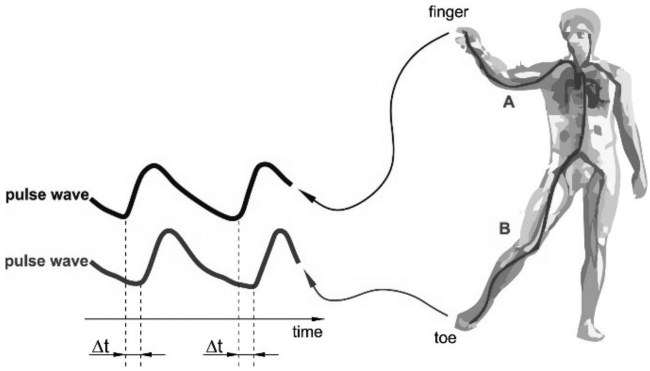


Fig. 7. Operating principle of pOpmètre; adapted from [96].

- *pOpmètre*® (Axelife SAS, France). The system is equipped with two photodiodes (Fig. 7), put on the finger and the toe [65], [96], [97], similar to pulse oximeters [98], in which the pulpar arteries are examined by an infrared (IR) beam, in order to measure the *finger-toe PWV* (ftPWV). The subject height is used to estimate the travelled distance, while the transit time Δt is measured between the pulse wave's feet at the finger and the toe. Such time interval approximates the aortic pulse transit time provided that the timings in the upper (A) and lower limbs (B) are comparable [96]. Among the advantages of using plethysmography, there are a good ease of use, rapidity, feasibility, movement artefacts reduction [27]. However, this technique is acceptable for the body peripheral sites only. Transit time results coming from pOpmètre and reference techniques agree, nevertheless it decreases for PWV assessment because of uncertainties on distance measurements [96]. In [88], the mean difference ± SD in PWV estimation between pOpmètre and the reference invasive method was -0.44 ± 4.44 m/s ($r = 0.41$), showing a weak relationship with invasive aortic outcomes.
- *PulsePen*® (DiaTecne, Italy). The system simultaneously or sequentially detects the pressure waveforms at carotid and femoral arteries, by means of applanation tonometry. It is equipped with two small portable tonometers (pressure probes) and a small ECG unit [99]. The travelled distance is assessed through the subtractive method using a tape-measure placed on the tonometer probe. Transit time is determined at the wave foot through the intersecting tangent method [100]. In [88], the mean difference ± SD of 0.20 ± 2.54 m/s ($r = 0.71$) in PWV estimation between PulsePen-ETT and the reference invasive method (i.e., FS-Stiffcath catheter) has been found.
- *SphygmoCor*® (AtCor Medical, Australia). The device allows sequential arterial pulse recording at the carotid and femoral sites through applanation tonometry [101], gating the two signals using the R-waves of a simultaneously recorded ECG (*SphygmoCor* CvMS) [66], [102]. Quality controls are needed to prevent whatever detrimental effect due to heart rate variability on the PWV evaluation. A new version of the device (*SphygmoCor* XCEL) allows to acquire simultaneous pressure waves through a cuff and

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tonometer system [102], [103], reducing the measurement time by slightly more than 50% [104]. In both the cases, transit time between the waveforms is estimated applying the foot-to-foot method (intersecting tangent algorithms). The main difference between the two devices is related to the different distance measured value despite the common subtractive method used [102]. In [88], the mean difference $\pm$ SD in PWV estimation between SphygmoCor and the reference invasive method (i.e., FS-Stiffcath catheter) was -0.61 ± 2.57 m/s ($r = 0.70$).

- *Ultrasound systems*® Ultrasound devices are today the only ones that allow to perform the non-invasive measurement of the elastic properties of the arterial wall (e.g., Young modulus), as well as to evaluate the relationship between such properties and intima-media thickness [1]. They allow not only the direct visualization and assessment of the vessel distention (e.g., by means of B-Mode and M-Mode), but also the measurement of the blood flow by Doppler Imaging. The main differences among the more common ultrasound techniques is the way in which the PWV assessment is carried out. B-Mode and M-Mode can measure the local PWV, while Doppler is used for the regional PWV estimation at several sites, although the carotid and femoral ones are often used [1], [105]. As in (3), the PWV can be estimated through two main parameters: the aortic area or diameter variation and the pressure variation. The latter can be measured by both by B-Mode or M-Mode tracing of the ascending or the descending aorta. Since aortic distensibility, measured by the variation of the cross-sectional area, showed better correlation with PWV than aortic distensibility measured by the variation of the aortic diameter, the B-Mode is preferable [106] despite a worsening of the frame rate [49]. An improvement in precision in the measurement of diameter (for both diastole and systole phase), up to 10 times compared to the video-image analysers, has been obtained with the development of echotracking devices (e.g., Philips *HDI-lab*, Aloka *E-Traking*, Esaote *RF-QAS*). Indeed, the precision in determining the diameter stroke variation seems to be around 1 μm for echotracking systems and 150 μm (i.e., pixel size) for video-image analysers [1]. On the other hand, ultrasound Doppler records flow waveforms from accessible sites (e.g., carotid and femoral sites) from which PWV can be estimated. Waveforms can be acquired both in an ECG gating sequential and in a simultaneous way [77], [107]. Distance is measured through the subtractive method, using the suprasternal notch. The flow wave foot from each recording site allows to determine the time interval [78]. Ultrasound Doppler is used to measure aoPWV with reliability and reproducibility providing similar results to Complior [77]. Furthermore, B-Mode ultrasound displays an anatomical image increasing the measurements precision. In [107], a good agreement between invasive and Doppler measurement of PWV is found with a mean difference $\pm$ SD of 0.13 ± 0.79 m/s.
- *VP1000*® – Vascular Profile – (Omron Healthcare, Japan). The device is based on 4 oscillometric cuffs placed on both arms (brachial) and ankles to simultaneously record brachial and post-tibial arterial pressure waveforms [108] and calculate baPWV [78]. The travelled distance is determined through the automatic Weber method [56], [75], while the transit time is derived from the time delay between the foot of the proximal and distal waveforms. The wave foot is found as the starting point of the sharp systolic upstroke [56], [59], [65]. baPWV measurement includes a much longer pathway, with the assumption that the pressure waves transit time values for both upper and lower limbs are similar.

Other commercial systems are proposed in the market, as summarized in Table IV. In summary, the PWV measurements of the majority of devices analysed in this work, whose values, were retrieved from several validation studies reviewed in [109], are listed in Table V.

VIII. Main Limitations of Available Devices

The main drawback preventing wide implementation of cf-PWV measurement in clinical practice, lies in the fact that a common accepted reference value and a general measurement protocol for travel distance is still not available [41] and the best way to assess it from surface measurements remains an outstanding issue [1], [75]. The curvatures within the vasculature are not considered in the external measurement of the distance. This limits the correct PWV assessment, especially in the presence of tortuous vessels [94]. Furthermore, it is necessary to consider that vessels decrease in size as they move away from the heart: peripheral vessels are narrower and thinner walled. The transit time, instead, is affected by the fact that arteries such as carotid, iliac and femoral ones, have not the same stiffness. In general, the accuracy in the measurement of the actual aoPWV value may decrease because in the case of one step exam execution, the pulse waveform propagates in opposite directions with respect to the heart (e.g., upstream in the carotid artery, downstream in the femoral one), while in the case of two steps exam execution, the beat-to-beat variability of the isovolumetric contraction time during ECG gating cannot be neglected [43], [79]. In the present section, the main limitations of each device analyzed in the previous section are reported and summarized in Table VI. The reported evaluations have been assigned by the Authors on the basis of the scientific literature analysis. As regards portability, the evaluation has been made according to the weight w of each device: not applicable if $w \geq 500$ kg; low if 50 kg $\leq w < 500$ kg; medium if 1 kg $\leq w < 50$ kg; good if 0.4 kg $\leq w < 1$ kg, very good if $w < 0.4$ kg.

In order to get accurate measurements with Arteriograph or other similar devices (e.g., Mobil-O-Graph), the cuff must be carefully positioned and very tightly wrapped around the arm. A further limitation is that the measurement is negatively affected by arm motion, therefore requiring the patient to stay still [79]. On the other hand, the size of the arm may prevent from good quality acquisitions of the brachial pulse wave (i.e., very thin or

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TABLE V
EXAMPLES OF PWV VALUES AND POPULATION SAMPLE AGE FROM VALIDATION STUDIES [109]

System		PWV [$m \cdot s^{-1}$]		Age
		Mean value	Standard deviation	
Arteriograph®		8.0	2.2	48 ± 15
Complior®		8.6	2.6	47 ± 16
Mobil-O-Graph®		8.5	2.1	69 ± 11
MRI		6.5	1.1	59 ± 10
pOpmètre®		8.8	2.4	53 ± 20
PulsePen®		8.2	3.5	51 ± 20
SphygmoCor®	CvMS	8.7	2.2	60 ± 13
	XCEL	7.0	1.7	46 ± 20
US Doppler systems		10.0	3.1	59 ± 16

TABLE VI
OVERVIEW OF THE MAIN LIMITATIONS FOR THE REVIEWED COMMERCIAL DEVICES

System		Portability	User independence	Movement artefact independence	Ergonomics and patient comfortability	Distance measurement accuracy	PWV measurement accuracy	Reference
Arteriograph®		●●●●	●●●	●●●	●●●	●●	●●	[43,79,81,84, 109]
Complior®		●●●	●	●	●	●●	●●	[88,89,109,111]
MRI		○	○	●	●●●	●●●●	●●●●	[62,109]
pOpmètre®		●●●	●●●●	●●●●	●●●●	●	●	[88,96,97,109]
PulsePen®		●●●●	●	●	●	●●	●●	[88,99,100,109]
SphygmoCor®	CvMS	●●●	●	●	●	●●	●	[88,101,103, 104,109,112]
	XCEL	●●●	●●	●●	●●		●●	
US systems	B- M- Mode	●	●	●	●	●●●	●●●	[1,77,107,109]
	Doppler					●●		
VP1000®		●●	●●●	●●●	●●●	●●	●	[56,108]

Portability = device small in size and weight, powered by batteries; *User independence* = the device can be used by a single person, without expert assistance; *Movement artefact independence* = device insensitive to motion and sensors positioning disturbances; *Ergonomics and patient comfortability* = device suited to the psychophysical needs of workers and comfortable for the patient; *Distance measurement accuracy* = accuracy of the estimation method used to measure the pulse travel distance; *PWV measurement accuracy* = accuracy of the PWV estimation with respect to the invasive assessment. The judgments reported have been assigned by the Authors on the basis of the scientific literature analysis. ●: Low; ●●: Medium; ●●●: Good; ●●●●: Very good; ○: Not applicable.

very thick arms), especially for low brachial pressure values; in such a case, the detection of the reflection point on the recorded waveform may be compromised [110].

Although Complior system is believed to be the gold-standard in carotid-femoral PWV measurement [77], it is affected by some limitations [111], including: operator-skill dependence in the positioning of sensors, difference between the actual path length of the pulse wave and the distance measured between the transducers, high sensitivity of the piezoelectric to motion and positioning disturbances and low comfort of the system (some patients experimented as the clip on their neck is not comfortable).

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The extent to which ftPWV, measured by pOpmètre, refers to central or peripheral AS is still unknown and a lower sensibility with respect to the cfPWV has been found [96]. Moreover, the biophysical principles underlying the functioning of the arterial tree should be deeply studied through detailed biophysical models, to clarify the agreement level between ftPWV and cfPWV. Indeed, the impossibility of using anatomical references of pulse measurements complicates the distance estimation, which makes use of the subject's height. One of the main issues concerning the drawbacks of PulsePen is related to the choice of measuring PWV with a single probe (through fast successive measurements) or with two probes (simultaneous recordings) [100]. Furthermore, the device requires skilled operator/s and the measurement is still hard to perform while the patient is standing.

SphygmoCor CvMS may be affected by the left ventricle contractility change or the heart rate variability because the measurements are made in sequence. Moreover, the sequential pulse acquisition is time-consuming. The system does not allow the production high quality recordings of radial and femoral pulse waveforms, therefore in patients with lower blood pressure, peripheral arterial disease or with obesity, peripheral pulses can be more difficult to record with respect to healthy subjects [112]. As in Complior, this system is quite intrusive and a high degree of operator expertise in the peripheral signal acquisition is required [104]. On the other hand, SphygmoCor XCEL device has become more independent from the operator through the cuff-based acquisition protocol as well as user friendly maintaining its accuracy. Although tonometry acquisition of the peripheral pulse captures much higher-frequency information, a compensation occurs as this is inherently damped by cuff volumetric displacement [104].

Most of ultrasound systems use video-image analysis and this results in a limitation of the measurements precision [1]. The frequency of the PWV signal is so high that the frame rate of a common US medical system is usually not enough to perform a correct wave sampling on a small segment of the aorta without aliasing. To overcome this, a very large segment of the aorta is usually considered, and ad hoc devices are designed and developed, mostly for applied clinical research. The development of automatic methods for the transit time estimation is recommended to lower the variability due to manual calipers in Ultrasound Doppler [77].

Finally, the baPWV from VP1000 by means of standardized cuffs reduces the operator dependence on the measurement results, nevertheless baPWV is an index of peripheral arterial stiffness only [56], [113].

As a final consideration, starting from well-established practical experiences collected in time [1], that resulted in the release of several devices in the market, novel solutions based on non-contact measurements have been proposed in literature [27] in the last years. Among them, optical techniques should be taken into account because they allow PWV monitoring without direct contact with the subject [114]–[116]. On the other hand, the scientific research is moving towards feasibility and validation studies aiming at monitoring and preventing CVD: for example, a technical route to solve existing challenges is represented by small and cheap wearable sensors based on different technologies, such as photoplethysmography (PPG) [117], [118] or microelectromechanical systems (MEMS) [119].

IX. Conclusion

Arterial stiffness is an important predictor of CVD or future CV events (i.e., any incidents that may cause damage to the heart muscle). The increasing costs of CVD has motivated the development of different devices able to estimate predictive parameters strictly linked to the arterial health state. Despite the wide range of current commercially available devices for PWV assessment, the main detrimental limitations are related to the pulse travel distance measurement. This issue is due to the necessity of a long arterial pathway which results in the estimation of a regional PWV rather than local aoPWV. Indeed, even if cfPWV is accepted as the gold-standard for non-invasive AS measurement, it revealed itself as a mere surrogate. Nowadays, there is no general agreement on how to measure travel distance in a non-invasive way. Consequently, a lack in reference values occurs in clinical practice for non-invasive cfPWV estimation. Among the reviewed devices available in the market, the ones based on tonometric/piezoelectric technology are usually affected by operator skills as well as by motion artefacts. Even if devices embedded with pressure cuffs are more user friendly than others, since the cuff inflation causes a dampening of higher-frequency information in some cases they provide an index of peripheral AS rather than central AS.

In general, it may be asserted that, in currently available technology for PWV measurements, there is no device whose characteristics are non-invasiveness and comfort for the patient, as well as high measurement accuracy, suitability for local estimation of the PWV, user friendly and low dependence on operator. In conclusion, although there is a high interest in arterial stiffness measurements and many technical solutions to PWV evaluation are proposed in the market as well in literature, novel methods and protocols should be studied and developed to overcome the drawbacks and limitations of today available devices. These devices should be aimed at reducing the length of the investigated arterial path, should provide accurate measurements and have a better ergonomics.

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